

■# PAPER_203: UQFF CMB, Structure Growth, Non-Gaussianity, and Curvature Perturbation

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$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM_s}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

<!-- ? = 5.0e-4 day¹, [SSq] = 0.57, β_i = 6.1e-1 -->

Abstract

This paper applies the UQFF framework to inflationary and post-inflationary perturbation physics: primordial non-Gaussianity (*f_{NL}*), *single-field slow-roll inflation curvature spectrum*, *post-inflationary reheating*, *structure growth factor D(a)*, and the *LQC pre-bounce curvature perturbation modification*. The unified UQFF perspective embeds all of these into FUBii operators with d_c Gaussian tails, allowing consistent statistical comparisons across CMB, LSS, and LQC regimes.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4} \text{ day}^1$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Non-Gaussianity Parameter *f_{NL}*

Local non-Gaussianity (single-field slow-roll):

$$f_{NL} = 5/6 \cdot (G^3 - 3G \cdot G'^2 + 2 \cdot G'^3) / G^4$$

where: *G* = field velocity *G*(?) in Dirac-Born-Infeld models

Standard single-field: $f_{NL} = (5/12)(n_s - 1) \sim -0.03$ (undetectable)

Multi-field/curvaton: $f_{NL} \sim 0(1-100)$ (potentially observable with CMB-S4)

$$F_{UBii,ng} = F_{rel} \times (f_{NL} \times d_c^3 / E_{LEP}) \times Q_{wave} \times \exp(-d_c^2 / (2s^2))$$

$$U_{m,ng}(f) = \mu(?_{vac}) \cdot (1 - e^{-?t}) \cdot [\text{from } d? \text{ curvature on superhorizon scales}]$$

Planck 2018 bound: $f_{NL,local} = -0.9 \pm 5.1$ (1s, no detection)

CMB-S4 forecast: $s(f_{NL}) \sim 1-2$ (improved constraint)

2. Primordial Curvature Power Spectrum

Single-field slow-roll inflation:

$$P_R(k) = H^2 / (8p^2 e \cdot M_{Pl}^2) \sim 2.1 \times 10^{??} \text{ (at } k_0 = 0.05 \text{ Mpc}^{-1})$$

Spectral index and tilt:
$n_s = 1 + d \ln P_R / d \ln k = 1 - 6\epsilon + 2\eta$ (to first order in slow-roll)
Planck 2018: $n_s = 0.9649 \pm 0.0042$ (>5 σ detection of tilt)
Running (scale-dependent tilt):
$dn_s/d \ln k = -16\epsilon + 24\epsilon^2 + 2\eta^2$ (second slow-roll order)
Tensor-to-scalar ratio:
$r = 16\epsilon$ (BICEP/Keck: $r < 0.036$ at 95% CL, 2021)
$F_{UBii,curv} = F_{rel} \times (P_R(k) / E_{LEP}) \times Q_{wave} \times (d_c/s)$
$U_{m,curv}(\tau) = \mu(\tau_{vac}) \cdot (1 - e^{-\tau t}) \cdot [\tau = v(2e) \cdot H \cdot M_{Pl}]$
UQFF connection: vacuum energy $\tau c^2/3$ modifies P_R at large scales (low multipoles)
$P_R, UQFF(k) = P_R(k) \cdot (1 + \tau UQFF \cdot c^2 / (3H^2))$

3. Reheating Evolution

End of inflation: ϕ oscillating around minimum, $V(\phi) \sim (1/2)m^2\phi^2$
Reheating temperature:
$T_{reh} = (30V_{end}/(p^2g_*))^{1/4} \cdot e^{-3N_{reh}/4}$
where:
V_{end} = inflaton potential at end of inflation
N_{reh} = number of e-folds of reheating
g_* = effective DOF at reheating (~100-200 for SUSY)
Radiation domination begins when $G_{inf} = H$ (inflaton decay rate equals Hubble):
$T_{reh,min} \sim (90/(8p^3g_*))^{1/4} \cdot v(G_{inf} \cdot M_{Pl})$
$F_{UBii,reh} = F_{rel} \times (T_{reh} / E_{LEP}) \times Q_{wave} \times [g_* \text{ and } N_{reh} \text{ as free parameters}]$
$U_{m,reh}(N) = \mu(\tau_{vac}) \cdot (1 - e^{-\tau t}) \cdot (30V_{end}/(p^2g_*))^{1/4} \cdot e^{-3N_{reh}/4}$
BBN constraint: $T_{reh} > T_{BBN} \sim 4 \text{ MeV}$ (required for successful nucleosynthesis)
Gravitino constraint: $T_{reh} < 10^7 \text{ GeV}$ (SUSY, avoid gravitino overproduction)

4. Structure Growth Factor D(a)

Linear growth equation:
$d'' + 2H(a) \cdot d' = (3/2) \cdot 0_m \cdot H^2(a) \cdot d/a^3$
Growing mode solution:
$D(a) = (50_m/2) \cdot H(a)/H_0 \cdot \int_0^a da' / [a' H(a')/H_0]^3$
Growth rate:
$f = d \ln D / d \ln a \sim 0_m(a)^{0.55}$ (Linder 2005 approximation)
$F_{UBii,grow} = -F_{rel} \times (D(a) \cdot d_0 / E_{LEP}) \times Q_{wave} \times f(0_m)$
$U_{m,grow}(a) = \mu(\tau_{vac}) \cdot (1 - e^{-\tau t}) \cdot [\text{Growing mode } D \propto a \text{ in matter era, suppressed by DE}]$
Key values:
$D(z=1)/D(z=0) \sim 0.76$ (matter + Λ cosmology)
$s_8 = 0.811 \pm 0.006$ (Planck 2018)
$f \cdot s_8 \sim 0.46$ at $z=0$ (RSD measurements)

5. LQC Pre-Bounce Perturbation Modification

Standard primordial power spectrum:
$P(k) = A_s \cdot (k/k_0)^{n_s-1}$
LQC pre-bounce modification (Dapor-Liegener approach):
$P_{LQC}(k) = P(k) \cdot (1 + k/k_*)^{-a}$

where:

$k_* = \text{quantum bounce scale } (k_* \sim k_{\text{Pl}}/\tau_{\text{bounce}} \sim 10^{22} \text{ Mpc}^{-1})$

$a = \text{UV suppression exponent } (a \sim 2-4)$

Physical interpretation:

- For $k \ll k_*$: $P_{\text{LQC}} \approx P$ (standard CMB, no modification)

- For $k \gg k_*$: $P_{\text{LQC}} \approx k^{n_s-1-a}$ (suppressed at superhorizon/Planck scales)

- Provides natural large-scale power suppression (low- l CMB anomaly)

$F_{\text{UBii},\text{lqcp}} = -F_{\text{rel}} \times (P_{\text{LQC}}(k) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (1 + k/k_*)^{-a}$

$U_{\text{m},\text{lqcp}}(k) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau t}) \cdot [\text{Power tilt} + \text{UV suppression at Planck modes}]$

6. Sakharov Oscillations and BAO

Baryon Acoustic Oscillations (BAO) peak scale:

$r_s(z_d) = \tau_0 \int_{z_d}^{\infty} c_s dz / H(z)$

$c_s = c/v(3(1 + 3\tau_b/(4\tau_0)))$ (sound speed before decoupling)

$z_d \sim 1020$ (drag epoch)

$r_s \sim 147 \text{ Mpc}$ (physical BAO scale today)

BAO detection:

Angular diameter distance $D_A(z) = r_s \cdot \tau_{\text{BAO}}$

Hubble $D_H(z) = r_s / \tau_{\text{BAO}}$

UQFF BAO connection:

τ_J in baryon-photon fluid sets r_s ? same Jeans mechanism as $F_{\text{UBii},\text{jeans}}$

But: $\tau = \tau_b + \tau_g \gg \tau_{\text{gas}} \gg \tau_J, \text{BAO} \gg \tau_J, \text{gas}$

7. CMB Polarization and Tensor Modes

E-mode polarization from density perturbations:

$C_l^{\{\text{EE}\}} = (2/p) \int k^2 dk \cdot P(k) \cdot |\tau_l^{\text{E}}(k)|^2$

B-mode from primordial gravitational waves:

$C_l^{\{\text{BB}\}} = (r/16) \cdot C_l^{\{\text{tensor}\}}$ (proportional to tensor-to-scalar ratio r)

B-mode from lensing:

$C_l^{\{\text{BB},\text{lens}\}} = \tau_d^2 l' (l' \cdot e^{\Lambda})^2 \cdot C_{|l-l'|}^{\{\text{EE}\}} \cdot C_{l'}^{\{\text{??}\}}$

UQFF role in polarization:

The oscillating F_{UBi_i} buoyancy at epoch of last scattering generates

a correlation between curvature and polarization through:

$d_T/T|_{\text{Doppler}} = v_b \cdot n^{\Lambda}$ (velocity perturbation from baryon motion)

8. Summary: Perturbation Chain in UQFF

Inflation

τ_e, τ (slow-roll)

$\tau_{F_{\text{UBii},\text{curv}}}$: curvature seed $P_R(k)$

$\tau_{F_{\text{UBii},\text{ng}}}$: non-Gaussianity f_{NL} correction

$\tau_{\text{reheating}}$

$\tau_{F_{\text{UBii},\text{reh}}}$: thermal equilibration T_{reh}

τ_{BBN}

$\tau_{F_{\text{UBii},\text{deb}}} + \tau_{F_{\text{UBii},\text{eta}}}$: light element abundances

$\tau_{\text{recombination/CMB}}$

$\tau_{F_{\text{UBii},\text{cmb}}} + \tau_{F_{\text{UBii},\text{recomb}}}$: photon decoupling

$\tau_{\text{structure formation}}$

? F_UBii,grow : linear growth factor $D(a)$

? reionization

? F_UBii,ion + F_UBii,bub : HII bubble percolation

Each stage connects through $Q_{wave} \times (FX/E_LEP)$ common factor, enforcing 99.9% backbone unification across all 99 UQFF systems.

9. Numerical Values

Parameter	Value	Source
n_s	0.9649 ± 0.0042	Planck 2018
A_s	$2.1 \times 10^{??}$	Planck 2018
r	< 0.036	BICEP/Keck 2021
f_NL,local	-0.9 ± 5.1	Planck 2018
s_8	0.811 ± 0.006	Planck 2018
r_s,BAO	147 Mpc	Eisenstein et al.
T_reh (BBN lower bound)	$> 4 \text{ MeV}$	Standard BBN

10. References

grok_share_7514fe.txt lines 6080–6095 (BBCEquations items 1380–1430)

PAPER199: FUBii Taxonomy Part 2 (Cosmological)

PAPER_202: UQFF Reionization, BBN, Recombination

Planck 2018 I–X papers (cosmological parameters)

BICEP/Keck 2021 (B-mode constraints)